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EXAMINER	
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ART UNIT	PAPER NUMBER
3662	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 16/337,906	Applicant(s) Wiesner et al.	
	Examiner MAHMOUD M KAZIMI	Art Unit 3662	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 07/22/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-5,7-10,12,14-16 and 18 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-5,7-10,12,14-16 and 18 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/28/2019 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of Claims

This communication is in response to applicant's filing dated 07/22/2022. Claims 6, 11, 13 and 17 are canceled. Claims 1-5, 7-10, 12, 14-16 and 18 are currently pending.

Continued Examination Under 37 CFR 1.114

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 07/22/2022 has been entered.

Examiner's Note

Upon Applicant's amendments made in the Request for Continued examination, Examiner conducted an updated search for the newly added limitations and upon consideration prior art was found for the independent claims. Therefore, an art rejection was made in this Non-Final office action set below. The indication of allowability set

forth in the previous action is withdrawn and prosecution is reopened in view of the following new ground of rejection.

Response to Argument

Applicant's arguments filed on 07/22/2022, with respect to the previous 35 U.S.C. 101 rejection of claim 1 has been fully considered and is unpersuasive.

With respect to the previous 35 U.S.C. 101 rejection of claim 1, Applicant's arguments with respect to 35 USC § 101 directed to non-statutory subject matter been fully considered but they are not persuasive.

Examiner respectfully disagrees. Claims 1-5, 7-10, 12, 14-16 and 18 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a judicial exception (abstract idea) without significantly more. Examiner incorporates herein the response to arguments from the previous office actions. The proposed amendments do not overcome the 35 U.S.C. 101 rejection. Applicant amended the claims to include "... responsive to a user request received via an interaction with an interface, generating data based on at least on some of the aggregate profiles for providing a parking search route to a user". The concept is still directed to generating parking related data. Examiner addressed all the claim limitations in light of the new 2019 Patent Eligibility Guidance (2019 PEG).

The instant claims do not attempt to solve an unconventional technological solution, but rather use the processing circuitry as a tool to implement the abstract idea. The focus of the claimed invention in the present is not on an improvement in

computers as tools, but on certain independently abstract ideas that use computers as tools.

Under the 2019 PEG, Step 2A, prong two, applicant submits that the claims integrate the alleged judicial exception into a practical application. Examiner respectfully disagrees. integration into a practical application requires an additional element(s) or a combination of additional elements in the claim to apply, rely on, or use the judicial exception in a manner that imposes a meaningful limit on the judicial exception, such that the claim is more than a drafting effort designed to monopolize the exception. Limitations that are not indicative of integration into a practical application are those that are mere instructions to implement an abstract idea on a computer, or merely uses a computer as a tool to perform an abstract idea. -see MPEP 2106.05(f).

A claim may be found to be eligible if it integrates a judicial exception into a practical application as cited by Applicant. However, examiner notes that "claiming the improved efficiency inherent with applying the abstract idea on a computer" does not provide an inventive concept (see MPEP §2106.05(f)(2).) Claiming improved data processing efficiency inherent with applying any improvement to the judicial exception itself on a computer does not provide an inventive concept. The claims do not integrate the judicial exception into a practical application.

Examiner identified the focus of the invention, and determined that it is directed to an abstract idea and the use conventional “off the shelf” computer to implement the abstract idea.

To integrate the exception into a practical application, the additional claim elements must, for example, improve the functioning of a computer or any other

technology or technical field (see MPEP § 2106.05(a)), apply the judicial exception with a particular machine (see MPEP § 2106.05(b)), affect a transformation or reduction of a particular article to a different state or thing (see MPEP § 2106.05(c)), or apply or use the judicial exception in some other meaningful way beyond generally linking the use of the judicial exception to a particular technological environment (see MPEP § 2106.05(e)). See 2019 Revised Guidance.

None of the additional limitations is sufficient to integrate the judicial exception. The present claims do not disclose a technical solution to a technical problem. The claims here do not recite an improvement for electronic devices. Broadly disclosing obtaining positional data, analyzing the positional data and generating data based on at least on some of the aggregate profiles for providing a parking search route to a user, as in claim 1, does not improve the functioning of the computer, make it operate more efficiently, or solve any technological problem. Rather, the claim simply "includes instructions to implement an abstract idea on a computer" and "does no more than generally link the use of a judicial exception to a particular technological environment or field of use." 2019 Revised Guidance, 84 Fed. Reg. at 55. The claim uses generic computer components and generic computer functionality to generate parking related data. Claim 1 merely uses instructions to implement the abstract idea on a computer or, alternatively, merely uses a computer as a tool to perform the abstract idea. Here, the additional limitations do not integrate the judicial exception into a practical application.

The courts found that "... if a patent's recitation of a computer amounts to a mere instruction to 'implement[t]' an abstract idea 'on . . . a computer,' that addition cannot

impart patent eligibility." Alice Corp., 134 S.Ct. at 2358. The claimed invention does not indicate that specialized computer hardware is necessary to implement the claimed systems, similar to the claims at issue in Alice Corp. See Alice Corp., 134 S.Ct. at 2360 (determining that the hardware recited in the claims was "purely functional and generic," and did not "offer [] a meaningful limitation beyond generally linking the use of the [method] to a particular technological environment, that is, implementation via computers").

Applicant's claims in the instant application are similar to *Electric Power Group* because the claims disclose obtaining positional data..., analyzing the positional data... and generating data based on at least on some of the aggregate profiles for providing a parking search route to a user.

The claims here are not directed to a specific improvement to computer functionality nor an inventive solution to any computer specific problem. Also, limiting the use of an abstract idea "'to a particular technological environment' does not confer patent eligibility as this cannot be considered an improvement to computer or technology and so cannot be "significantly more."

Examiner notes that the processor limitations and the claim as a whole do not add significantly more than the abstract idea itself, because the claim does not amount to an improvement to the functioning of a computer itself; and the claim does not move beyond a general link of the use of an abstract idea to a particular technological environment. A generic recitation of a processor/device performing its generic computer functions does not make the claims less abstract.

The Court gave examples, which included an improvement to another technology or technical field; improvement to the function of the computer itself; or some other meaningful limitation beyond generally linking the use of an abstract idea to a particular technological environment. Such as in Diamond v. Diehr, the claims were found statutory in which the Arrhenius equation is used to improve a process of *controlling the operation of a mold in curing rubber parts*.

Examiner submits that under the current 35 USC 101 examining practice, the existence of such novel features would still not cure the deficiencies with respect to the abstract idea. See for example: *Ultramercial, Inc. v. Hulu, LLC, 112 USPQ2d 1750, U.S. Court of Appeals Federal Circuit, No. 2010-1544, Decided November 14, 2014, 2014 BL 320546, 772 F.3d 709, Page 1754 last two ¶*: "We do not agree with Ultramercial that the addition of merely novel or non-routine components to the claimed idea necessarily turns an abstraction into something concrete". Indeed, in this in instant case, the limitations simply narrow or limit the abstract idea without providing anything significantly more than the abstract idea itself.

For these reasons the rejection under 35 USC § 101 directed to non-statutory subject matter set forth in this office action is maintained.

Claim Rejections - 35 USC § 101

7. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

8. **Claims 1-5, 7-10, 12, 14-16 and 18** are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more.

Under Step 2A – Prong 1:

Claim 1, 16 and 18 recite the abstract idea concept of a method (claim 1), system (claim 16), and non-transitory computer readable medium (claim 18) for generating parking related data within a geographic network. This abstract idea is described in at least claims 1, 16 and 18 by, **analysing, for each one of the plurality of trips, the positional data to identify a parking space search portion of the trip; analysing the positional data corresponding to the parking search portions of the trips to obtain, for each one of at least one set of a plurality of different time periods, a relative value of the parking related parameter associated with each one of a plurality of segments of the map in respect of the time period, the parking related parameter being a time dependent parameter; clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter; and using, for each group, relative parking related parameter data associated with each of the segments of the group to obtain an aggregate profile indicative of the time dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group,** are considered mental process steps. The identified claim limitations that recites an abstract idea fall within the enumerated

groupings of abstract ideas in Section 1 of the 2019 Revised Patent Subject Matter Eligibility Guidance published in the Federal Register (84 FR 50) on January 7, 2019.

The limitations of **analysing, for each one of the plurality of trips, the positional data**, and **analysing the positional data corresponding to the parking space, clustering, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, using, the aggregate profile being applicable to each of the segments of the group** as drafted, are process that, under its broadest reasonable interpretation, covers performance of the limitation in the mind but for the recitation of generic computer components. That is, other than reciting "A method for generating parking related data" nothing in the claim element precludes the step from practically being performed in the mind. If a claim limitation, under its broadest reasonable interpretation, covers performance of the limitation in the mind but for the recitation of generic computer components, then it falls within the "Mental Processes" grouping of abstract ideas. Accordingly, the claim 1 recites an abstract idea.

Under Step 2A – Prong 2:

The claims recite elements additional to the abstract idea. However, these additional elements fail to integrate into a practical application.

Claim 1 recites, **obtaining positional data relating to a plurality of trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state and electronic map** and **navigable network and electronic map.** Which is mere data gathering that is simply employed as a tool to collect information and display results, which is

insignificant extra solution activity as the step simply gathers data necessary to perform the abstract idea. These additional steps amount necessary data gathering and generic computer components wherein all uses of the recited abstract idea require such data gathering or data output. See MPEP 2106.05(g).

Under Step 2B:

Regarding Step 2B of the 2019 PEG, independent claim 1 does not include additional elements (considered both individually or in combination) that are sufficient to amount to significantly more than the judicial exception for the same reasons to those discussed above with respect to determining that the claim does not integrate the abstract idea into a practical application. As discussed above with respect to integration of the abstract idea into a practical application, the additional elements amount to nothing more than applying the exception using a generic computer component. Generally applying an exception using a generic computer component cannot provide an inventive concept. And as discussed above, the additional limitations of **obtaining positional data relating to a plurality of trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state and electronic map** and **navigable network and electronic map**, **responsive to a user request received via an interaction, generating data based on at least on some of the aggregate profiles for providing a parking search route to a user**, the examiner submits that these limitations are insignificant extra-solution activities.

Further, a conclusion that an additional element is insignificant extra-solution activity in Step 2A should be re-evaluated in Step 2B to determine if they are more than

what is well-understood routine and conventional activity in the field. The additional limitations of obtaining positional data relating to a plurality of trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state and electronic map and navigable network and electronic map, are well-understood, routine and conventional activities because the background recites that obtaining steps are done by a GPS/GNSS receiver and an electronic map (or mathematical graph, as it is sometimes known), in its simplest form, is effectively a database containing data representative of nodes, most commonly representative of road intersections, and lines between those nodes representing the roads between those intersections. In more detailed digital maps, lines may be divided into segments defined by a start node and end node" and that the navigable network is simply a road network. Further the mere collection of data or receipt of data over a network is a well-understood, routine and conventional function when it is claimed in a merely generic manner (as it is here). See MPEP 2106.05(d). Therefore, claims 1, 16 and 18 are ineligible under 35 U.S.C. 101.

Dependent claims 2-5, 7-10, 12, 14, and 15 include limitations of claim 1 and only further defines mental process of claim 1. These further limitations are considered mental process without any significantly more elements to the abstract idea.

Claims 1-5, 7-10, 12, 14-16 and 18 are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claim(s) 1-3, 7-10, 12, 14-16 and 18 are rejected under 35 U.S.C. 103 as being unpatentable over Kandal, US Patent No. 8423275B1, in view of Eyal Amir, US 20120161986 A1, hereinafter referred to as Kandal and Amir, respectively.

Regarding claim 1, Kandal discloses a method for generating parking related data in respect of one or more navigable elements of a navigable network within a geographic area, each navigable element being represented by a segment of an electronic map, the electronic map comprising a plurality of segments representative of the navigable network, said comprising:

obtaining positional data relating to trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state (*The procedure has resource to motion profiles of users of navigation devices in the area that are radiolocation-aided, especially satellite-aided – See at least column 2 lines 61-62. For each user, a parking place found event is*

determined if he has found a parking place. The criteria for this can be a cessation of motion for a time that exceeds a fixed tolerance period and/or the navigation device having been positional removed from a street – See at least column 3 lines 7-12);

analysing, for each one of the plurality of trips, the positional data to identify a parking space search portion of the trip ("*the motion profile of the user in the area is assessed according to the invention-specific criteria only if the user is perceived as searching for a parking place.*" – *See at least column 4 lines 58-80. "A user especially can be perceived as searching for a parking place if he is in motion in the area over a time span that is above a preset minimum time at a speed below a minimum speed.*" – *See at least column 4 lines 83-88. "make a determination that the user is involved in a search for a parking place – See at least column 5 lines 15-18*); and

analysing the positional data corresponding to the parking space search portions of the trips to obtain, for each one of a plurality of segments of the electronic map, a parking related parameter associated with the segment *("make a determination that the user is involved in a search for a parking place, and thus for triggering a designation and evaluation of the motion profile" – See at least column 5 lines 15-18*).

Kandal fails to explicitly disclose clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter; using, for each group, relative parking related parameter data associated

with each of the segments of the group to obtain an aggregate profile indicative of the time dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group; and responsive to a user request received via an interaction with an interface, generating data based on at least on some of the aggregate profiles for providing a parking search route to a user.

However, Ami teaches:

clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter (*Steps performed by the facility in order to construct segment availability models in some embodiments. Among segments having adequate segment observations, the facility generates segment clusters. In some embodiments, a segment has adequate segment observations if at least four observations are contained for the segment by the segment availability observations table – See at least ¶48. for each class of segments, the facility constructs an availability model. In some embodiments, the facility constructs a polynomial regression model for each class of segments. In some embodiments the facility constructs a Bayesian network model for each class of segments – See at least ¶52. Additionally, it incorporates time of day, day of week, and holidays into its models – See at least ¶53 and FIG. 9-10*);

using, for each group, relative parking related parameter data associated with each of the segments of the group to obtain an aggregate profile indicative of the time

dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group (*For each such segment, the facility fits a polynomial of degree 3 to total number of available spots in the segment as a function of time of day. In various embodiments, the facility establishes a different set of clusters, classifications, and corresponding models for variations of various kinds, such as different time-of-day ranges, and/or different days-of-week or days-of-week ranges, i.e. time dependence – See at least 48. he facility proceeds to aggregate these determined parking times for each neighborhood, using such aggregation functions as mean, median – See at least ¶68*); and

responsive to a user request received via an interaction with an interface, generating data based on at least on some of the aggregate profiles for providing a parking search route to a user (*User interface diagrams showing sample displays presented by the facility in a parker client in some embodiments as part of providing a recommended route for searching for parking. The display includes a map showing the user's current location. The user can press the preferences button in order to set certain parking preferences, select a destination on the map, then press a park me button in order to request a recommended route for searching for parking near the destination – See at least ¶56 and FIGS 13A-13C*).

Kandal discloses a procedure for determining the probability of finding a parking place within a certain period of time on a pre-set stretch of road of an area with a multiplicity of stretches of road. Amir teaches providing guidance for locating street parking for a user.

It would have been obvious to one ordinary skill in the art before the effective filing date of the claimed invention to modify the invention of Kandal and include the feature of clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter, as taught by Amir, to guide a user along a route to seek a parking place in the vicinity of a travel destination.

Regarding claim 2, Kandal discloses wherein the parking space search portion is detected based on one or more criteria using the positional data in respect of the trip, the criteria including the positional data being indicative of one or more of; the vehicle performing one or more loops (*According to the invention, in the navigation device an endless loop route is computed along a section plan around the destination, and the user is guided along this loop until he has found a parking place – See at least column 5 lines 12-13 and column 6 lines 17-20*).

Regarding claim 3, Kandal discloses analysing the positional data in respect of the trip to identify a start of a parking space search during the trip, the start of the parking space search defining the start of the parking space search portion of the trip, wherein the step of identifying the start of a parking space search comprises analysing the positional data relating to a trip to identify a point at which the speed of travel of the vehicle is below a predetermined threshold, optionally wherein the predetermined threshold is set with respect to a free flow speed for the segment (*As a further criterion,*

or an alternative one, for the user of the navigation device being in search for a parking place, what can serve is detection that he has passed over a stretch of road at least for a second time. For often, searches for parking places are accompanied by multiple trips over a stretch of street or road in the hope that a parking place has been freed up in the meantime. Naturally additional criteria can be applied to make a determination that the user is involved in a search for a parking place, and thus for triggering a designation and evaluation of the motion profile – See at least column 5 lines 10-11, 15-18).

Regarding claim 7, Kandal discloses wherein the parking probability obtained for a segment is based on a ratio of the number of times the segment is traversed without finding a parking space according to the parking space search portion data to the number of times a parking space is successfully found on the segment according to the parking space search portion data (*For example, one minute can be assumed as the pre-set time period, and the figures for the users who have traveled along this stretch of road searching for parking places, are set in relation to those who within the pre-set one minute of travel time on this stretch of road have found a parking place. This relation then represents the probability. In the computation, still other criteria may be influential – See at least column 3 lines 55-60*).

Regarding claim 8, Kandal discloses wherein the parking related parameter is an expected parking search time (*One minute can be assumed as the pre-set time period, and the figures for the users who have traveled along this stretch of road searching for parking places, are set in relation to those who within the pre-set one*

minute of travel time on this stretch of road have found a parking place –See at least column 3 lines 55-60).

Regarding claim 9, Kandal discloses wherein the expected parking space search time is based on the average remaining time spent searching for a parking space after traversing the segment according to the parking space search portion data (*An especially simple display, and one that is bold for the user, results when a total of three probability groups are formed, with higher-probability values in a first probability group, average-probability values in a second probability group, and lower-probability values in a third probability group – See at least column 5 lines 50-55*).

Regarding claim 10, Kandal discloses wherein the parking related parameter is a time dependent parameter, and a plurality of values of the parameter are determined for each segment, each value of the parking related parameter being in respect of a different time period, preferably wherein the time periods are recurring time periods (*A further advantageous embodiment of the invention consists in having the day of the week and/or time of day on which or at which the "parking place found" event occurs be detected in addition to the data previously mentioned herein. For often the parking place situation in an area is strongly dependent on the time of day and also on the day of the week. In residential areas in which a large number of commuters reside, who during typical work times have departed with their cars, at these times of day it is often much easier to find parking places than in the evening, when residents have returned from*

their places of work and occupy existing parking places with their vehicles – See at least column 4 lines 25-37 and column 6 lines 6-13).

Regarding claim 12, Kandal discloses associating data indicative of the determined parking probability for the segment with each segment in the electronic map for which such data is determined (*Thus for example when a user has reached his destination or is approaching it, he can order a corresponding menu item on his navigation device which then, in the displayed map, graphically depicts the probability groups for finding parking places for the individual stretches of road. Depiction of these probability groups can also, however, be made to depend on a scaling of the displayed map. If an especially large and detailed display, thus a large scale, is used, then the probability groups can also be displayed – See at least column 5 lines 38-42*).

Regarding claim 14, Kandal discloses associating data indicative of the aggregate profile in association with each segment of the group (*Thus for example when a user has reached his destination or is approaching it, he can order a corresponding menu item on his navigation device which then, in the displayed map, graphically depicts the probability groups for finding parking places for the individual stretches of road. Depiction of these probability groups can also, however, be made to depend on a scaling of the displayed map. If an especially large and detailed display, thus a large scale, is used, then the probability groups can also be displayed – See at least column 5 lines 38-42*).

Regarding claim 15, Kandal discloses using the parking related parameter data obtained in generating a parking search route or in determining an estimated time to find a parking space (*An additional aspect of the invention is a procedure for displaying the probability of finding a parking place on a certain stretch of road, on a radiolocation-aided, especially satellite-aided, navigation device having digitally stored mapping data and a display device for displaying the mapping data – See at least column 5 lines 19-24*).

Regarding claim 16, Kandal discloses a system constructed and arranged to generate parking related data in respect of one or more navigable elements of a navigable network within a geographic area, each navigable element being represented by a segment of an electronic map, the electronic map comprising a plurality of segments representative of the navigable network, said system comprising:

obtain positional data relating to trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state (*The procedure has resource to motion profiles of users of navigation devices in the area that are radiolocation-aided, especially satellite-aided – See at least column 2 lines 61-62. For each user, a parking place found event is determined if he has found a parking place. The criteria for this can be a cessation of motion for a time that exceeds a fixed tolerance period and/or the navigation device having been positional removed from a street – See at least column 3 lines 7-12*);

analyse, for each one of the plurality of trips, the positional data in respect of the trip to identify a parking space search portion of the trip (*"the motion profile of the user in the area is assessed according to the invention-specific criteria only if the user is perceived as searching for a parking place." – See at least column 4 lines 58-80. "A user especially can be perceived as searching for a parking place if he is in motion in the area over a time span that is above a preset minimum time at a speed below a minimum speed." – See at least column 4 lines 83-88. "make a determination that the user is involved in a search for a parking place – See at least column 5 lines 15-18*); and

analyse the positional data corresponding to the parking space search portions of the trips to obtain, for each one of a plurality of segments of the electronic map, a parking related parameter associated with the segment *("make a determination that the user is involved in a search for a parking place, and thus for triggering a designation and evaluation of the motion profile" – See at least column 5 lines 15-18*).

Kandal fails to explicitly disclose cluster, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter; use, for each group, relative parking related parameter data associated with each of the segments of the group to obtain an aggregate profile indicative of the time dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group; and responsive to a user request

received via an interaction with an interface, generating data based on at least on some of the aggregate profiles for providing a parking search route to a user.

However, Ami teaches:

cluster, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter (*Steps performed by the facility in order to construct segment availability models in some embodiments. Among segments having adequate segment observations, the facility generates segment clusters. In some embodiments, a segment has adequate segment observations if at least four observations are contained for the segment by the segment availability observations table – See at least ¶48. for each class of segments, the facility constructs an availability model. In some embodiments, the facility constructs a polynomial regression model for each class of segments. In some embodiments the facility constructs a Bayesian network model for each class of segments – See at least ¶52. Additionally, it incorporates time of day, day of week, and holidays into its models – See at least ¶53 and FIG. 9-10*);

use, for each group, relative parking related parameter data associated with each of the segments of the group to obtain an aggregate profile indicative of the time dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group (*For each such segment, the facility fits a polynomial of degree 3 to total number of available spots in the segment as*

a function of time of day. In various embodiments, the facility establishes a different set of clusters, classifications, and corresponding models for variations of various kinds, such as different time-of-day ranges, and/or different days-of-week or days-of-week ranges, i.e. time dependence – See at least 48. he facility proceeds to aggregate these determined parking times for each neighborhood, using such aggregation functions as mean, median – See at least ¶68); and

responsive to a user request received via an interaction with an interface, generating data based on at least on some of the aggregate profiles for providing a parking search route to a user (*User interface diagrams showing sample displays presented by the facility in a parker client in some embodiments as part of providing a recommended route for searching for parking. The display includes a map showing the user's current location. The user can press the preferences button in order to set certain parking preferences, select a destination on the map, then press a park me button in order to request a recommended route for searching for parking near the destination – See at least ¶56 and FIGS 13A-13C*).

Kandal discloses a procedure for determining the probability of finding a parking place within a certain period of time on a pre-set stretch of road of an area with a multiplicity of stretches of road. Amir teaches providing guidance for locating street parking for a user.

It would have been obvious to one ordinary skill in the art before the effective filing date of the claimed invention to modify the invention of Kandal and include the feature of clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the

segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter, as taught by Amir, to guide a user along a route to seek a parking place in the vicinity of a travel destination.

Regarding claim 18, Kandal discloses a non-transitory computer readable medium having stored thereon a computer program product comprising instructions which, when read by a computing device, cause the computing device to operate according to the method for generating parking related data in respect of one or more navigable elements of a navigable network within a geographic area, each navigable element being represented by a segment of an electronic map, the electronic map comprising a plurality of segments representative of the navigable network, said method comprising:

obtaining positional data relating to trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state (*The procedure has resource to motion profiles of users of navigation devices in the area that are radiolocation-aided, especially satellite-aided – See at least column 2 lines 61-62. For each user, a parking place found event is determined if he has found a parking place. The criteria for this can be a cessation of motion for a time that exceeds a fixed tolerance period and/or the navigation device having been positional removed from a street – See at least column 3 lines 7-12*);

analysing, for each one of the plurality of trips, the positional data to identify a parking space search portion of the trip ("*the motion profile of the user in the area is assessed according to the invention-specific criteria only if the user is perceived as*

searching for a parking place." – See at least column 4 lines 58-80. "A user especially can be perceived as searching for a parking place if he is in motion in the area over a time span that is above a preset minimum time at a speed below a minimum speed." – See at least column 4 lines 83-88. "make a determination that the user is involved in a search for a parking place – See at least column 5 lines 15-18); and

analysing the positional data corresponding to the parking space search portions of the trips to obtain, for each one of a plurality of segments of the electronic map, a parking related parameter associated with the segment *("make a determination that the user is involved in a search for a parking place, and thus for triggering a designation and evaluation of the motion profile" – See at least column 5 lines 15-18*).

Kandal fails to explicitly disclose clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter; using, for each group, relative parking related parameter data associated with each of the segments of the group to obtain an aggregate profile indicative of the time dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group; and responsive to a user request received via an interaction with an interface, generating data based on at least on some of the aggregate profiles for providing a parking search route to a user.

However, Ami teaches:

clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter (*Steps performed by the facility in order to construct segment availability models in some embodiments. Among segments having adequate segment observations, the facility generates segment clusters. In some embodiments, a segment has adequate segment observations if at least four observations are contained for the segment by the segment availability observations table – See at least ¶48. for each class of segments, the facility constructs an availability model. In some embodiments, the facility constructs a polynomial regression model for each class of segments. In some embodiments the facility constructs a Bayesian network model for each class of segments – See at least ¶52. Additionally, it incorporates time of day, day of week, and holidays into its models – See at least ¶53 and FIG. 9-10*);

using, for each group, relative parking related parameter data associated with each of the segments of the group to obtain an aggregate profile indicative of the time dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group (*For each such segment, the facility fits a polynomial of degree 3 to total number of available spots in the segment as a function of time of day. In various embodiments, the facility establishes a different set of clusters, classifications, and corresponding models for variations of various kinds, such as different time-of-day ranges, and/or different days-of-week or days-of-week*

ranges, i.e. time dependence – See at least 48. he facility proceeds to aggregate these determined parking times for each neighborhood, using such aggregation functions as mean, median – See at least ¶68); and

responsive to a user request received via an interaction with an interface, generating data based on at least on some of the aggregate profiles for providing a parking search route to a user (*User interface diagrams showing sample displays presented by the facility in a parker client in some embodiments as part of providing a recommended route for searching for parking. The display includes a map showing the user's current location. The user can press the preferences button in order to set certain parking preferences, select a destination on the map, then press a park me button in order to request a recommended route for searching for parking near the destination – See at least ¶56 and FIGS 13A-13C*).

Kandal discloses a procedure for determining the probability of finding a parking place within a certain period of time on a pre-set stretch of road of an area with a multiplicity of stretches of road. Amir teaches providing guidance for locating street parking for a user.

It would have been obvious to one ordinary skill in the art before the effective filing date of the claimed invention to modify the invention of Kandal and include the feature of clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar

relative values of the parking related parameter, as taught by Amir, to guide a user along a route to seek a parking place in the vicinity of a travel destination.

1. **Claims 4 and 5** are rejected under 35 U.S.C. 103 as being unpatentable over Kandal, US Patent No. 8423275B1, in view of Eyal Amir, US 20120161986 A1, as applied to claim 1 above and further in view of Levine et al., US 20100052946 A1, hereinafter referred to as Kandal, Amir and Levine, respectively.

Regarding claim 4, the combination of Kandal and Amir fails to explicitly disclose wherein said identified point is considered to be a candidate point for the start of a parking space search, the method comprising considering one or more further criteria to determine whether the candidate point may be confirmed as the start of a parking space search, wherein the criteria include one or more of; the point being within a predetermined distance of the end of the trip, the point being within a predetermined distance of a turn performed by the vehicle and/or being within a loop performed by the vehicle.

However, Levine teaches considering one or more further criteria to determine whether the candidate point may be confirmed as the start of a parking space search, wherein the criteria include one or more of; the point being within a predetermined distance of the end of the trip, the point being within a predetermined distance of a turn performed by the vehicle and/or being within a loop performed by the vehicle (*In some embodiments of the present invention, application server 120 may recognize that the user is in proximity to his destination and thus, for example, the period of time from arriving to the destination area, for example when the distance from the destination gets lower than a certain threshold until a full stop of the travel, which may be interpreted as*

parking, may be identified by application server 120 as the search for a parking spot – See at least ¶22).

Kandal discloses a procedure for determining the probability of finding a parking place within a certain period of time on a pre-set stretch of road of an area with a multiplicity of stretches of road. Amir teaches providing guidance for locating street parking for a user. Levine teaches a system and method for parking time estimations, the system and method may provide to a user rich picture of the availability of parking spaces in an area.

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the combination of Kandal and Amir and include the feature of to determine whether the candidate point may be confirmed as the start of a parking space search, wherein the criteria include one or more of; the point being within a predetermined distance of the end of the trip, the point being within a predetermined distance of a turn performed by the vehicle and/or being within a loop performed by the vehicle, as taught by Levine, to guide a user along a route to seek a parking place in the vicinity of a travel destination.

Regarding claim 5, the combination of Kandal and Amir fails to explicitly disclose wherein the step of determining whether the candidate point may be confirmed as the start of a parking space search further takes into account a distance travelled at the speed below the predetermined threshold.

However, Levine teaches wherein the step of determining whether the candidate point may be confirmed as the start of a parking space search further takes into account a distance travelled at the speed below the predetermined threshold (*when the distance*

from the destination gets lower than a certain threshold until a full stop of the travel, which may be interpreted as parking, may be identified by application server 120 as the search for a parking spot –See at least ¶22).

Kandal discloses a procedure for determining the probability of finding a parking place within a certain period of time on a pre-set stretch of road of an area with a multiplicity of stretches of road. Amir teaches providing guidance for locating street parking for a user. Levine teaches a system and method for parking time estimations, the system and method may provide to a user rich picture of the availability of parking spaces in an area.

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the combination of Kandal and Amir and include the feature of wherein the step of determining whether the candidate point may be confirmed as the start of a parking space search further takes into account a distance travelled at the speed below the predetermined threshold, as taught by Levine, to guide a user along a route to seek a parking place in the vicinity of a travel destination.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. **Wiesner et al. US 10720057 B2**.

Wiesner discloses obtaining profile data indicative of the time dependence of a time dependent parameter for association with one or more segments of an electronic

map, as well as a system, such as a server, on which part or all of the method may be implemented.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MAHMOUD M KAZIMI whose telephone number is (571)272-3436. The examiner can normally be reached M-F 7am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Aniss Chad can be reached on 571-270-3832. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/M.M.K./
Examiner, Art Unit 3662

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